1. Write short notes with necessary diagram if needed:
    a. Multipath Fading.
    b. Rayleigh Fading.
    c. Rician Fading.
    d. Fast Fading vs Slow Fading
    e. Okumara-hata Path loss Model
    f. Multiplexing
    g. Multiple Access
    h. Doppler Effect.
2. What is virtual carrier sensing? Explain the scenery that can be solved by this technique?
3. For radio transmission in free space, signal power is reduced in proportion to the square of the distance from the source, whereas in wire transmission, the attenuation is a fixed number of dB per kilometer. The following table is used to show the dB reduction relative to some references for free space radio and uniform wire. Fill in the missing numbers to complete the table.

| Distance (km) | Radio (dB) | Wire (db) |
|---|---|---|
| 1 | -6 | -3 |
| 2 | | |
| 4 | | |
| 8 | | |
| 16 | | |

4. Suppose a transmitter produces 50 W of power.
    a. Express the transmit power in units of dBm and dBW.
    b. If the transmitter's power is applied to a unity gain antenna with a 900-MHz carrier frequency, what is the received power in dBm at a free space distance of 100 m?
    c. Repeat (b) for a distance of 10 km.
    d. Repeat (c) but assume a receiver antenna gain of 2.
5. Explain how the following categories of noise are generated, and is there any strategy to handle these noises?
    a. Thermal noise
    b. Intermodulation Noise
    c. Crosstalk
    d. Impulse noise
6. Write down the architecture of 802.11 WLAN (BSS and IBSS).
7. Explain the DCF process.
8. Explain the necessity of different IFS.
9. Explain the PCF process.
10. What are access networks and core networks? Use a diagram to explain. Write down the taxonomy of Wireless access networks.

11. Explain the process of time domain backoff. What happened to backoff process if the ack is not received by sender? What is the disadvantages of time domain backoff?
12. Explain Back2F technique, which addresses the disadvantages of time domain backoff. Explain the necessity of two rounds of backoff in Back2F.
13. Explain the application and architecture of the following network:
    a. DTN
    b. Mesh Network
    c. Wireless Sensor Network
    d. VANET
    e. Low PAN